

KITTLA KENTTAROVA, Finland

WMO: 028240

Lat: 67.987N

Lon: 24.243E

Elev: 1147

StdP: 14.10

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.0	-9.4	-19.6	2.0	-14.9	-13.5	2.8	-9.3	23.0	24.9	21.4	23.7	3.6	340	0.746

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	75.2	60.4	70.9	58.4	66.8	56.6	62.9	71.3	60.3	68.2	58.2	64.4	7.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.9	80.3	65.2	57.2	72.8	62.0	55.3	67.9	60.1	28.9	71.5	27.2	68.9	25.7	64.4	69.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
18.9	16.6	14.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-20.9	79.0	4.3	5.0	-24.0	82.6	-26.6	85.5	-29.0	88.3	-32.2	91.9		
			-20.0	64.8	5.9	3.3	-24.2	67.1	-27.7	69.0	-31.0	70.9	-35.2	73.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.4	10.9	12.6	20.2	30.1	41.1	49.6	56.6	52.3	44.6	31.0	21.6	16.9		
	DBStd	17.52	10.56	11.89	8.13	6.86	8.30	7.29	6.98	6.30	6.05	6.92	8.31	9.53		
	HDD50	6884	1211	1048	925	596	302	94	17	46	178	589	853	1025		
	HDD65	11917	1676	1468	1390	1046	742	464	275	397	612	1054	1303	1490		
	CDD50	461	0	0	0	0	25	82	221	117	16	0	0	0		
	CDD65	21	0	0	0	0	1	2	15	3	0	0	0	0		
	CDH74	157	0	0	0	0	8	18	113	18	0	0	0	0		
	CDH80	16	0	0	0	0	0	0	14	2	0	0	0	0		
	WSAvg	7.0	6.5	7.2	8.3	7.2	7.1	7.4	6.6	6.6	7.0	6.7	6.6	7.4		
	PrecAvg	23.3	1.6	1.3	1.1	1.3	1.5	2.5	3.3	2.8	2.2	1.9	2.0	1.7		
(15) Precipitation	PrecMax	29.8	3.1	3.0	2.4	3.0	3.8	4.2	5.6	5.7	4.4	3.7	3.8	3.0		
	PrecMin	17.4	0.4	0.5	0.4	0.4	0.3	0.4	1.3	1.3	0.6	0.5	0.6	0.4		
	PrecStd	3.6	0.6	0.6	0.6	0.6	0.8	0.9	1.2	1.1	1.1	0.8	0.8	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.4	37.8	39.3	55.2	73.9	77.0	82.2	76.1	62.4	47.9	40.4	36.3		
		MCWB	32.1	35.9	34.4	43.8	55.6	61.9	64.4	64.1	53.6	44.6	39.0	34.4		
	2%	DB	29.9	32.8	36.1	47.2	65.7	71.2	77.1	71.0	58.6	45.4	36.8	33.0		
		MCWB	29.1	31.3	32.0	39.4	52.1	57.0	61.3	59.5	52.5	43.3	35.2	32.0		
	5%	DB	27.8	30.2	33.3	43.5	60.3	66.1	72.9	66.1	55.9	43.2	33.2	31.3		
		MCWB	27.2	29.4	30.7	36.7	49.1	54.4	59.7	57.4	51.3	41.8	32.7	30.7		
	10%	DB	25.2	27.8	31.2	40.2	54.0	61.8	68.5	62.3	53.5	40.2	31.6	29.5		
		MCWB	24.6	27.1	29.2	34.9	45.3	51.8	58.3	55.7	50.0	39.0	31.3	28.9		
	0.4%	WB	32.5	36.2	35.0	44.4	56.2	63.7	66.2	66.0	56.7	45.7	39.9	34.9		
		MCDB	33.3	38.2	38.3	53.8	69.6	72.8	74.5	73.8	59.7	46.5	40.5	36.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	29.4	31.4	32.9	39.9	53.2	59.7	63.6	61.1	54.2	43.8	35.3	32.4		
		MCDB	29.9	32.3	35.2	46.0	65.5	66.8	72.2	67.6	56.9	45.0	36.3	32.9		
	5%	WB	27.2	29.5	31.2	37.5	49.5	56.5	61.5	58.5	52.2	41.5	32.7	30.8		
		MCDB	27.7	30.3	33.0	42.8	58.7	63.0	70.1	64.1	55.0	42.9	33.1	31.4		
	10%	WB	24.5	27.0	29.4	35.5	46.4	53.7	59.6	56.6	50.5	39.2	31.4	29.0		
		MCDB	25.0	27.7	31.1	39.8	53.4	60.8	67.5	61.0	52.8	40.2	31.6	29.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	9.7	12.3	13.8	14.8	14.4	14.7	13.8	10.4	7.4	8.2	9.5		
		MCDBR	10.1	9.2	11.2	16.1	22.7	21.4	20.8	18.7	13.9	8.9	7.3	8.2		
		MCWBR	10.1	8.4	8.8	9.6	12.3	10.5	9.4	9.1	8.9	7.1	7.3	8.0		
		MCDBR	9.9	9.5	9.6	15.6	20.9	18.2	17.6	16.3	10.8	7.9	7.2	7.7		
	5% WB	MCWBR	10.0	8.8	8.0	9.7	11.9	10.3	9.3	8.9	7.9	6.8	7.2	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.184	0.244	0.280	0.310	0.329	0.332	0.341	0.326	0.297	0.263	0.350	N/A		
	taud		1.917	2.092	2.188	2.219	2.422	2.493	2.479	2.503	2.559	2.323	2.780	N/A		
	Ebn at Noon		36	177	240	265	275	277	270	262	243	175	52	0		
	Edh at Noon		3	15	25	32	30	29	28	25	18	12	4	0		
(44) All-Sky Solar Radiation	RadAvg		12	158	633	1245	1550	1577	1475	1066	563	187	30	0		
	RadStd		1	19	40	67	151	117	146	83	62	32	4	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page